Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store

A Proposal Report for BDM Capstone Project

Submitted by

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<u>Declaration Statement</u>

I am working on a Project titled " **Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store** ". I extend my appreciation to **Jhansi Rope Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: (**Digital Signature**)

Name: Pooja Sahu

Date: 10/07/25

1. Executive Summary and Title

Project Titled: “Business Data Management for a Dual Market Utility Goods Store: A Case Study on Jhansi Rope Store”

Jhansi Rope Store is a small, third-generation, family-run shop located in Subhash Ganj, Jhansi. The business supplies household utility products like ropes, mops, wipers, niwar, and twine, catering to both B2B and B2C customers across urban and rural regions.

The store faces challenges such as fluctuating seasonal demand, stockouts, overstocking, and difficulty identifying fast- and slow-moving products. These issues lead to poor stock planning, excess inventory, and ultimately, reduced profitability due to blocked capital and missed sales opportunities.

To address these challenges, we will analyze historical sales data using tools like charts and spreadsheets to uncover seasonal trends and customer demand patterns. The data will be structured into tables with key fields like product type, quantity, date of sale, and customer type. Through data aggregation, visualizations, and pattern identification, we aim to generate insights on demand cycles and inventory flow.

The expected outcome of the project is a data-driven inventory strategy that helps reduce excess stock, prevent stockouts, and improve overall operational efficiency. This will enable Jhansi Rope Store to make smarter stocking decisions, reduce money blockage in inventory, and ultimately enhance profitability while maintaining better product availability for its diverse customer base.

2. Organizational Background

Established in 1965, Jhansi Rope Store is a third-generation, family-run business located in Subhash Ganj, Jhansi. The store has long served as a trusted source for everyday utility products such as mops, wipers, ropes, niwar, and twine. It caters to both retail and wholesale customers across urban and rural regions, meeting the needs of households, small retailers, farmers, and service providers. The store is currently managed by Anshu Sahu, the grandson of the founder, who continues to uphold the legacy of reliability and community trust. By sourcing goods directly from manufacturers, Jhansi Rope Store ensures affordability, consistency, and timely availability. What began as a small neighborhood shop has grown into a well-known local business that’s deeply rooted in personal relationships and practical service. Through decades of honest work and commitment to customer satisfaction, the store has become a dependable name in utility product supply for the surrounding region.

3. Problem Statement

3.1 Inability to forecast seasonal demand accurately: Due to the absence of historical sales analysis and trend tracking, the store often overstocks or runs out of key products during seasonal shifts.
3.2 Lack of clarity on product movement: Without proper data, the business cannot distinguish between fast- and slow-moving items, leading to poor inventory decisions and capital being tied up in unsold stock.
3.3 **Manual inventory management systems:** Reliance on manual tracking limits visibility into real-time stock levels, slows down restocking, and reduces responsiveness to customer needs.
3.4 **External demand unpredictability:** Factors like festivals, weather changes, and rural vs. urban market variations are not accounted for in current planning, increasing the mismatch between stock and actual demand.

4. Background of Problem

Inability to forecast seasonal demand accurately: Jhansi Rope Store does not currently use historical sales data to predict when certain products will be in higher or lower demand. This leads to poor anticipation of customer needs during peak seasons like festivals, harvest periods, or school openings. As a result, the store often faces stock outs or is left with excess unsold inventory. This affects both sales performance and customer satisfaction.

Lack of clarity on fast- and slow-moving products: Without proper data tracking, the store is unable to identify which products sell quickly and which remain unsold for long periods. This creates challenges in optimizing shelf space and restocking decisions. Slow-moving items block working capital and storage space, while fast-moving items may run out when needed most. This imbalance contributes to overall inefficiency in operations.

Manual and inefficient inventory tracking system: The current inventory management process relies heavily on manual records and staff memory, which is prone to human error. This results in outdated stock counts, delayed restocking, and inconsistent supply. Lack of automation also makes it difficult to get a clear, real-time view of stock availability. These inefficiencies hinder the store's ability to respond quickly to customer demand.

External demand unpredictability: External factors such as festivals, weather conditions, and the differing needs of urban versus rural customers significantly influence product demand. However, these are not considered in the store's planning process due to a lack of structured data or predictive tools. As a result, the business cannot align its inventory with actual seasonal or regional demand. This leads to mismatched stock and lost sales opportunities.

5. **Problem Solving Approach**

To help Jhansi Rope Store overcome its inventory and demand-related challenges, we will follow a structured, data-driven approach that is both practical and easy to implement. Our goal is to turn existing sales and stock data into meaningful insights that improve decision-making and business performance.

- **Data Collection**
  We will begin by collecting past sales data including product names, dates of sale, quantities sold, and customer types (retail or wholesale). This will provide a base for understanding product movement and customer trends.

- **Data Cleaning and Structuring**
  The collected data will be organized in a consistent format using spreadsheets or a relational database. This ensures the data is easy to work with and suitable for analysis.

- **Descriptive Analysis**
  Using tools like charts and spreadsheets, we will analyze
  - Fast- and slow-moving products
  - Inventory issues such as overstocking or frequent stockouts
  - Seasonal sales trends to forecast demand

  This helps identify what's working and what needs adjustment.

- **External Demand Mapping**
  We will look at how external factors—such as festivals, weather patterns, and rural vs. urban consumer behavior—impact product demand. Aligning this context with sales data will help improve seasonal stock planning.

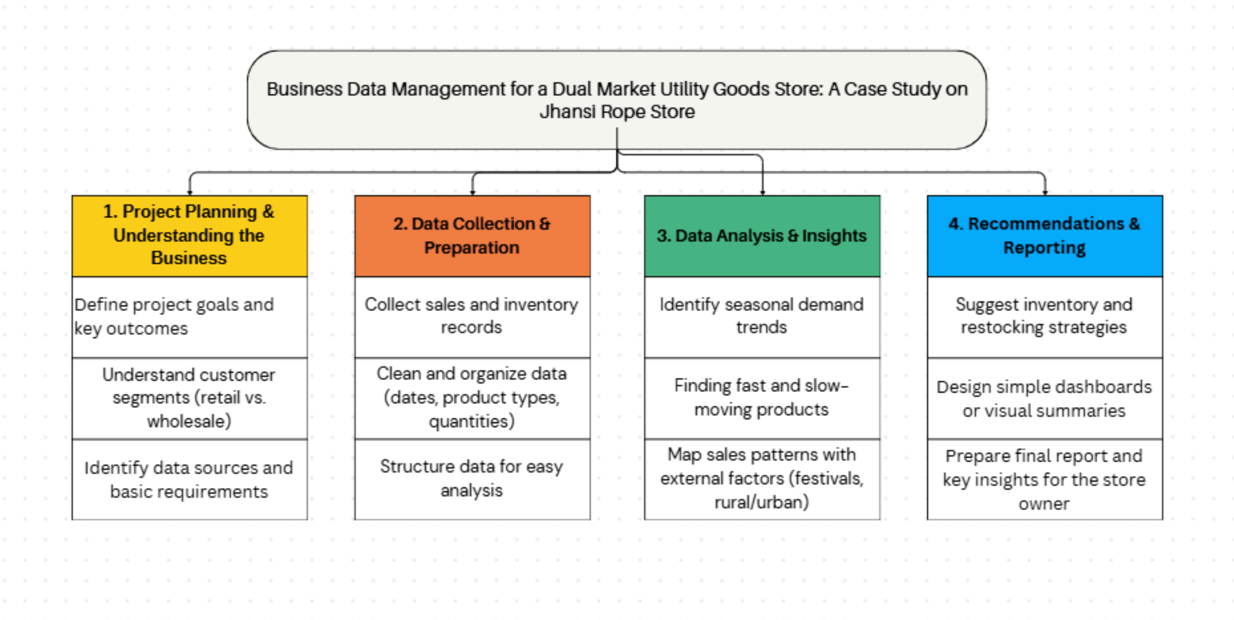
- **Visualization and Reporting**
  Charts and tables (line graphs, bar charts, pivot tables) will be used to present findings clearly, making it easy for the store owner to understand trends and take timely action.

- **Practical Recommendations**
  Based on the insights, we will suggest simple, effective steps such as:
  - When and how much to restock specific items
  - How to avoid unnecessary inventory buildup
  - How to align stock levels with real demand cycles

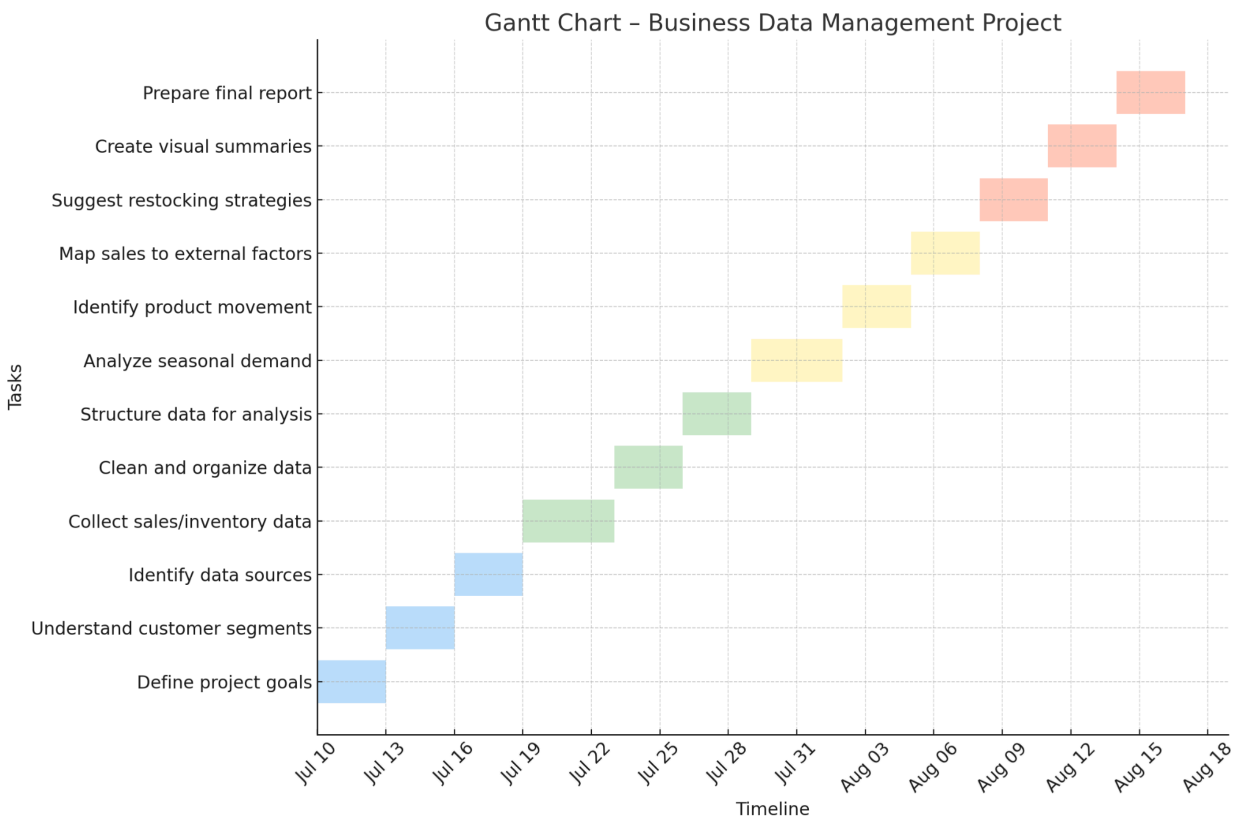
By following this approach, we aim to directly address all four project objectives—leading to smarter inventory decisions, better demand forecasting, reduced waste, and improved profitability for both B2B and B2C operations.

6. Expected Timeline

6.1 Work Breakdown Structure



6.2 Gantt Chart



7. Expected Outcomes

- **Clear seasonal demand insights**: Using past sales trends, the store will be able to better anticipate demand spikes (e.g., during festivals or harvest times), reducing both overstocking and running out of key products.

- **Visibility into product movement**: The project will highlight which items are fast-moving and which are not, so the store can prioritize high-demand goods and free up space from stagnant stock.

- **Organized and efficient inventory tracking**: By moving away from purely manual systems, the store will have a cleaner, more reliable way to track stock levels — helping it restock in time and reduce losses.

- **Better planning for external factors**: The analysis will also consider how weather, customer type (rural vs. urban), and local events affect demand — allowing the store to stock more intelligently.

- **Practical and actionable recommendations**: Rather than complex tools, the store will receive simple visual dashboards, charts, and a clear final report to guide future planning in both B2B and B2C operations.